PS-26

Emergency Supply Rover for Obstacle-Filled Environments

Hardware | Robotics, Mechatronics and Humanitarian Technology

NODE-X

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PROPOSED SOLUTION

Solution Overview

- RESQ-ROVER: a low-cost, fully autonomous 4-wheel rover that carries a first-aid kit across rubble to a drop zone.
- It scans, steers around obstacles, protects itself and unlocks the kit on arrival.

Key Features

- Fully autonomous waypoint driving
- 180° radar finds the widest gap
- Auto-recovery when it gets stuck
- Locked kit opens only at drop zone

How It Addresses the Problem

- **Navigate and avoid:** waypoints + radar
- **Failsafes:** tilt, stuck, low battery, link loss
- **Record:** route, events, delivery result

Unique / Innovative Aspects

- **Two controllers:** STM32 for reflexes, ESP32 for decisions
- **Live location** metric of the bot via dashboard
- Self-leveling cradle keeps the kit upright

SYSTEM DESIGN

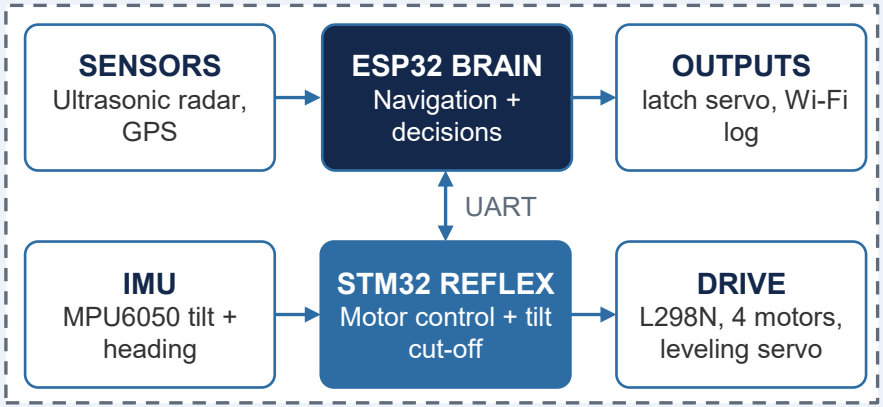
Proposed Methodology

1. Chassis and schematics design
2. Build and program the rover and integrating with dashboard
3. Testing using dashboard

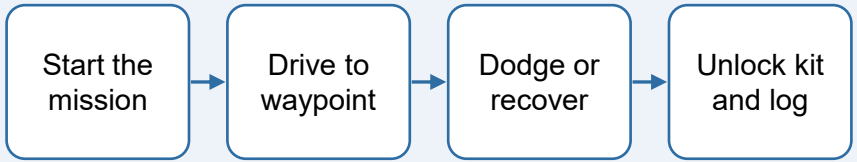
Working Principle

- **Sense:** radar, tilt, battery
- **Decide:** ESP32 picks the next action
- **Act:** STM32 drives motors, stops if tilt > 25°
- **If the link drops:** the mission continues onboard

Block Diagram / System Architecture



Flowchart / Workflow



Failsafes can interrupt at any step

IMPLEMENTATION AND TESTING

Technologies / Components Used

- **Software:** C++ firmware, OpenSCAD, Next.js dashboard
- **Hardware:** STM32, ESP32, L298N, 4 BO motors, HC-SR04 + servo, MPU6050, OLED, NEO-6M GPS

Testing Methodology

- 3 m × 3 m team-built obstacle course
- **Zones:** debris, sudden obstacle, ramp, dead-end, drop zone
- 5 runs per zone; pass = no collision

Implementation / Prototype Status

- **Done:** design, pin map
- **Next:** CAD parts, assemble, tune, run course tests
- **Finally:** demo video and mission log

Key Performance Metrics / Results

Metric	Target	Result
Collisions	0	Pending
Tilt cut-off	> 25°	Pending
Cruise speed	0.3 m/s	Pending
Delivery success	4 of 5 runs	Pending

BENEFITS & IMPACTS

Target Users / Beneficiaries

- Disaster response teams (NDRF, fire and rescue)
- NGOs, hospitals and trapped survivors

Real-World Applications

- Earthquake and landslide relief
- Reaching collapsed buildings and cut-off villages

Expected Impact

- Delivers medicine without risking rescuers
- Low-cost design from common, easy-to-find parts

Scalability / Future Potential

- Stronger motors and chassis for heavier loads
- Camera to spot survivors; rover fleets

THANK YOU

Questions and Discussion

Team NODE-X

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